Farzan Mohammadi

Istanbul, Turkey · +90 507-128-6650 · farzan.mohamadii@gmail.com
linkedin.com/in/farzan-mohammadi-669382131 · github.com/FarzanMohamadi · Google Scholar

Profile

Backend and software engineer with around ten years of experience building large-scale platforms, custom frameworks, and production systems — from ERPs and booking platforms to my own PHP MVC framework and ORM. A genuine curiosity for how things work beneath the surface pulled me into applied security and reverse engineering, where I bring the same builder's mindset to anti-bot, fingerprinting, and client-integrity systems, turning hard, undocumented problems into working tools. I also keep one foot in research, with two peer-reviewed publications on evaluating large language model reasoning. I'm happiest solving problems where the target keeps moving and the answer isn't written down anywhere.

Education

Khajeh Nasir Toosi University of Technology (KNTU) *2020 - 2023*
M.Sc. in Information Technology Engineering (E-commerce Engineering) Tehran, Iran

Shahroud University of Technology (SUT) *2015 - 2019*
B.Sc. in Computer Engineering Shahroud, Iran

Work Experience

Reverse Engineer & Web Engineer *Dec 2024 - Present*
BlackScale Media Cyprus - Remote

- Conducted applied security research on the anti-automation, client-integrity, and request-authentication defenses of large-scale web and native (Android) platforms.
- Deobfuscated large volumes of obfuscated, packed, and minified client-side JavaScript, reverse-engineering a major platform's dynamic module-loading and token-generation systems to recover their algorithms and request-construction logic.
- Reverse-engineered fully XHR- and GraphQL-driven web application flows, reconstructing request/response structures and undocumented query schemas.
- Built an automated parser that extracts request-construction logic from heavily bundled production JavaScript, and an automated dynamic parser that constructs and structures GraphQL queries from client bundles and simulates realistic end-to-end traffic.
- Investigated the correlation between HTTP/2 and TLS fingerprints (JA3/JA4-style) and worked to replicate Chrome's TLS fingerprint from scripted clients to avoid protocol-level detection.
- Bypassed SSL/TLS certificate pinning at the native (C/JNI) layer using Frida, hooking low-level socket and cryptographic functions rather than relying on Java-layer bypasses.
- Rooted and jailbroke Android devices to enable low-level dynamic instrumentation and study app-specific anti-tampering, integrity-verification, and root-detection behavior.
- Intercepted and reverse-engineered device/app attestation token generation and validation, including structural analysis of the underlying cryptographic schemes.
- Applied Frida to hook runtime functions and recover the logic of obfuscated client-side algorithms and verification routines.
- Analyzed client-side and native anti-analysis defenses, including transport-layer certificate pinning, and studied their detection and enforcement mechanisms.
- Investigated automated-challenge and verification subsystems (interaction-based checkpoints) to understand their design and analysis surfaces.
- Built a Chromium-based browser instrumentation harness to study browser fingerprinting surfaces and network-layer behavior under varied environment configurations.
- Developed custom interception and comparison tooling (Burp-style, in Java) to study and diff scripted vs. native client flows in detail, with dedicated logging to detect request/response mismatches and humanize traffic for both web and app.
- Reverse-engineered end-to-end request/response flows using Burp Suite and custom traffic-analysis tooling; decoded and documented undocumented request parameters and their integrity/verification logic, and designed an automated pipeline that parses protocol state and derives subsequent request steps programmatically.
- Used Selenium and Puppeteer for browser-automation-based reverse engineering and controlled traffic reproduction, and built a PHP-based traffic-replay platform to reproduce and validate reverse-engineered flows in a reproducible test environment.

Senior Programmer *2023 - Jan 2025*
SaadatRent Tehran, Iran

- Led development of a Laravel platform for an international car-rental company: dynamic booking website, mobile application, and CRM.
- Implemented database-level encryption with Libsodium and hardened handling of sensitive customer data.
- Designed a custom nested-set trait for hierarchical data in Laravel 10, with query optimization for large trees.

CTO and Senior Programmer *2018 - 2023*
Saman Ban Company Mashhad, Iran

- Designed and implemented a custom PHP MVC framework (routing engine and ORM) and built a large-scale web-based ERP with 20+ integrated modules (CRM, Supply Chain, HR, Finance, and more).
- Built a web-based property-management system and an instant-messaging tool integrated with the project-management and CRM software.
- Integrated IoT devices with the ERP system for secure device communication.

Backend Programmer *2022 - 2023*
Resalat Bank Tehran, Iran

- Led backend development of a social platform for bank customers.
- Built a PHP interface converting MySQL queries to Oracle SQL with a custom translation algorithm ($>$95% accuracy).
- Implemented end-to-end encryption for secure data transactions.

Web Developer (Remote) *2021 - 2022*
Mata Trading Company (via Samanban) Dubai, UAE

- Led end-to-end development and customization of a BPMS platform to streamline business operations, including user training and support.

Freelance Software Engineer *2014 - 2023*
Independent Remote

- Collaborated with HEC Montreal on a climate-data analysis project, producing geographical visualizations of US climate data in Python.
- Delivered 10+ full-stack web applications (online shops, exhibition sites) and internal automation for a major newspaper organization.
- Built a search engine over a local database in Java.

Publications

LogiCue: Targeted Prompting for Improved Modal and Conditional Reasoning in LLMs | Procedia Computer Science, Elsevier (ACLing 2025) *Dec 2025*

- Pattern-specific prompting framework for seven modal inference patterns, evaluated across nine models with statistical validation ($p < 0.01$).
- View publication

Can We Predict LLM Reasoning Failures? Structural Predictability of Modal and Conditional Inference Errors | KI 2026: 49th German Conference on Artificial Intelligence — Bremen, Germany *Aug 2026*

- Trained classifiers to predict per-instance LLM failures from input-side structural features (3,776 instances, nine models, three prompting strategies; AUC-ROC 0.69–0.93).
- Showed that targeted prompting and output-side reliability mechanisms are complementary.
- View publication

Certifications

IELTS (Overall Band 6.5) | IELTS Official — Credential ID 23IR002271MOHF120A *Jun 2023*

GRE General Test | ETS — Credential ID 86404796 (verify) *Nov 2023*

Technical Skills

Reverse Engineering: JavaScript deobfuscation, obfuscated/packed/minified code analysis, protocol & application reverse engineering, anti-bot / anti-automation analysis, browser fingerprinting internals, TLS/certificate-pinning analysis (JA3/JA4)

Security & Analysis: Frida (dynamic instrumentation), Burp Suite, traffic analysis, Puppeteer, Selenium, attestation & token-scheme analysis

Languages: JavaScript, Node.js, Python, PHP, Java, SQL

Frameworks & Tools: Laravel, RESTful APIs, WebSockets, RabbitMQ, Kafka, Redis, ElasticSearch, Git

Machine Learning: PyTorch, Scikit-learn, Pandas, NumPy, SciPy

Databases: MySQL, MariaDB, Oracle, PostgreSQL, MongoDB

Cloud & DevOps: AWS, Microsoft Azure, Google Cloud Platform, Docker, Kubernetes